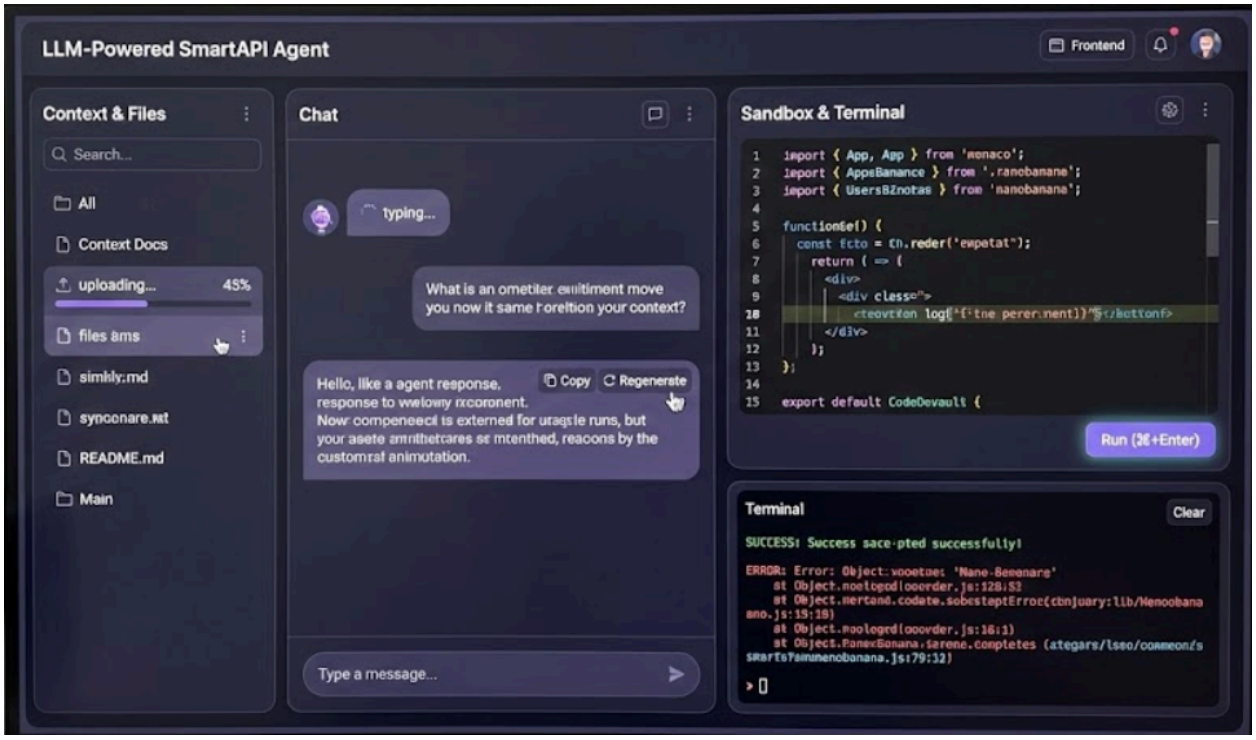
Frontend



Phase 1: Foundation & Styling Upgrade

- **Goal:** Set up a professional design system from day one.

1. **Initialize Project:** (Same as before) Vite + React + TypeScript.
2. **UI Component Library:** Instead of raw Tailwind, use a headless component library for accessible, unstyled primitives, and style them with Tailwind.
    - **Action:** Install **Shadcn UI** (built on Radix Primitives and Tailwind). This gives you beautifully designed, copy-paste components (buttons, inputs, dialogs, toasts) that are fully customizable.
3. **Theme & Typography:**
    - Define a refined dark theme palette (e.g., inspired by "Dracula" or "Nord" themes) with vibrant accent colors for primary actions.
    - Use a modern sans-serif font like **Inter** for UI text and a clean monospace font like **JetBrains Mono** or **Fira Code** for code.

Phase 2: Core Components with Enhanced Interactivity

- **Goal:** Build components that feel alive and responsive.

1. **Sidebar (Context & History):**
    - **File Uploader:** Add a visual **progress bar** using Shadcn's Progress component for uploads. Show success/error icons next to files.

    - **Search:** Add a search input at the top to filter past chats or uploaded documents.
2. **Chat Interface (The "Brain"):**
    - **Typing Indicators:** Display a "three-dot" animation bubble when waiting for the LLM's response.
    - **Message Actions:** Add small hover-actions on each agent message: a **"Copy"** icon for text/code and a **"Regenerate"** icon.
    - **Enhanced Markdown:** Style react-markdown output with typography plugins for better headers, lists, and tables.
3. **Code/Sandbox Panel (The "Lab"):**
    - **Editor:** Configure Monaco Editor with a vibrant theme (like "Dracula") and enable features like minimap, bracket matching, and line numbers.
    - **Toolbar:** Make the "Run" button prominent with a play icon and add a **keyboard shortcut hint** (e.g., "Run (⌘+Enter)"). Add a "Clear Terminal" button.

Phase 3: Integration & Dynamic Feedback

- **Goal:** Connect the frontend to the backend with real-time feedback.

1. **Streaming Responses:** Implement server-sent events (SSE) or WebSockets to stream the LLM's response token by token, making the chat feel faster and more conversational.
2. **Toast Notifications:** Use Shadcn's Sonner or Toast component to show non-intrusive alerts for actions like "File uploaded successfully," "Code copied to clipboard," or "Execution failed."

Phase 4: Polish, Animation & UX

- **Goal:** The final layer of professional sheen.

1. **Animations:** Use **Framer Motion** for subtle UI transitions:
    - Smoothly fade in new messages.
    - Animate the sidebar opening/closing.
    - Add a subtle pulse to the "Run" button when code is ready.
2. **Tooltips:** Add Shadcn Tooltip components to all icons (e.g., hover over the "Upload" icon to see "Upload API Docs (PDF, MD)").
3. **Loading States:** Replace spinners with sleek "skeleton" loaders that mimic the shape of the content being loaded (chat messages, file lists).